

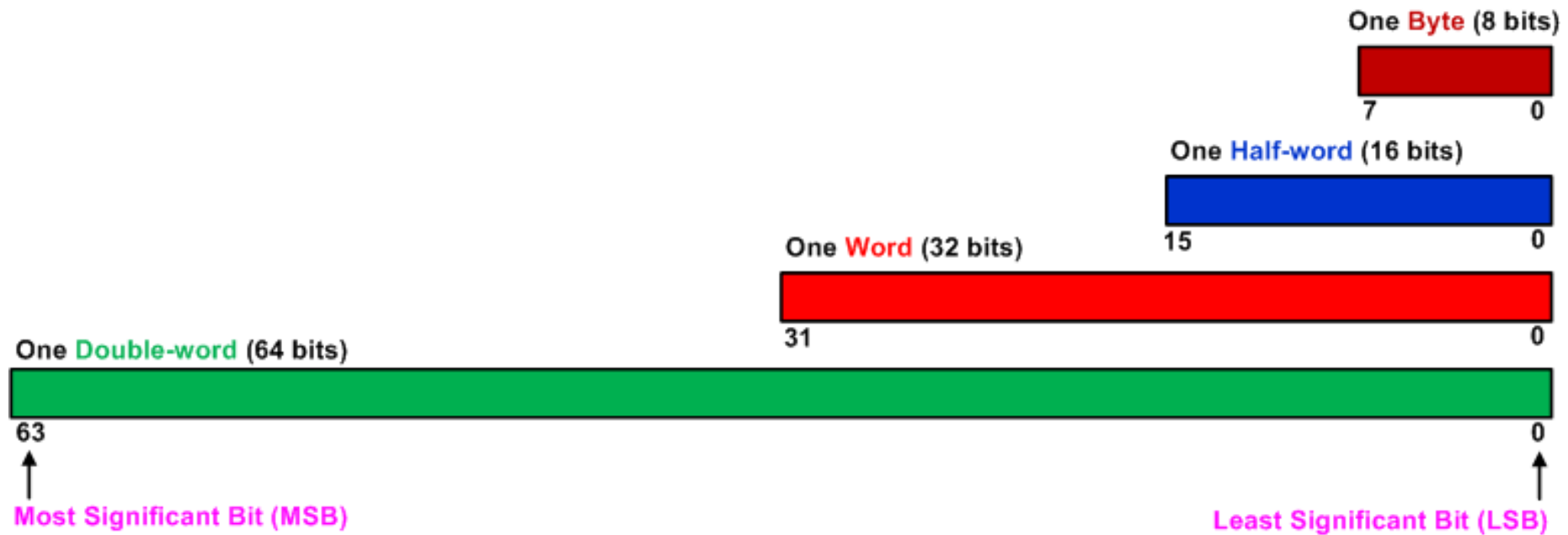
Embedded Systems with ARM Cortex-M Microcontrollers in Assembly Language and C

Chapter 2 Data Representation

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Bit, Byte, Half-word, Word, Double-Word

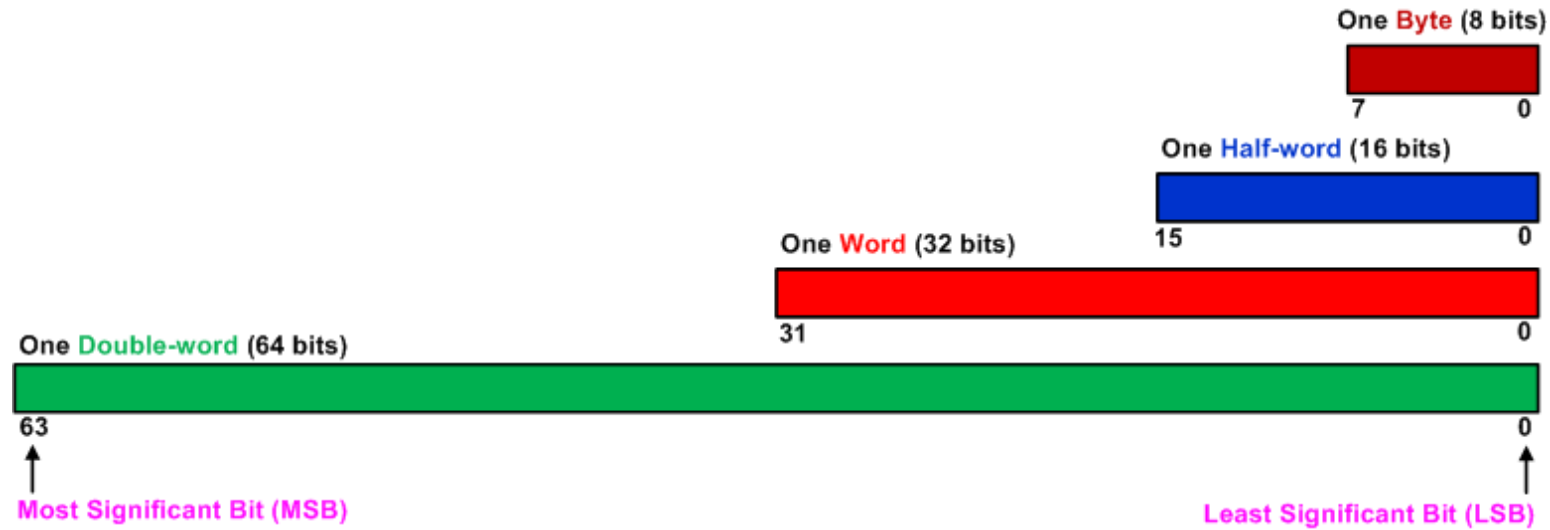


Binary, Decimal and Hex

Decimal	Binary	Hex
0	0000	0x0
1	0001	0x1
2	0010	0x2
3	0011	0x3
4	0100	0x4
5	0101	0x5
6	0110	0x6
7	0111	0x7
8	1000	0x8
9	1001	0x9
10	1010	0xA
11	1011	0xB
12	1100	0xC
13	1101	0xD
14	1110	0xE
15	1111	0xF

0x: Hex

Range of Unsigned Integers

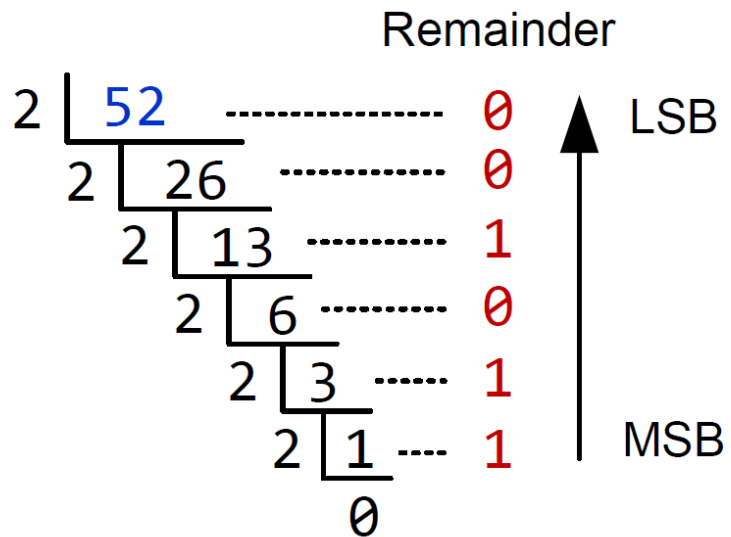


Storage Size	Range	Powers of 2
Unsigned Byte	0 to 255	0 to 2^8-1
Unsigned Half-word	0 to 65,535	0 to $2^{16}-1$
Unsigned Word	0 to 4,294,967,295	0 to $2^{32}-1$
Unsigned Double-word	0 to 18,446,744,073,709,551,615	0 to $2^{64}-1$

Unsigned Integers

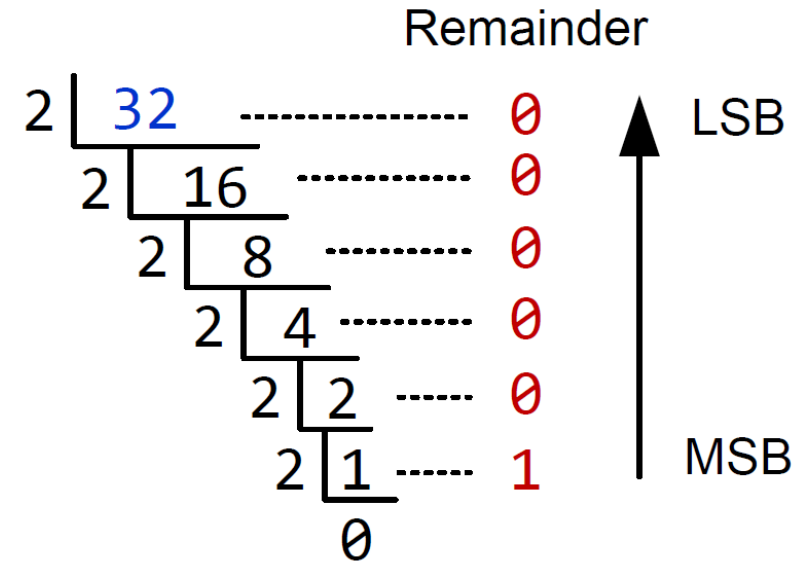
Convert Decimal to Binary

Example 1



$$52_{10} = 110100_2$$

Example 2



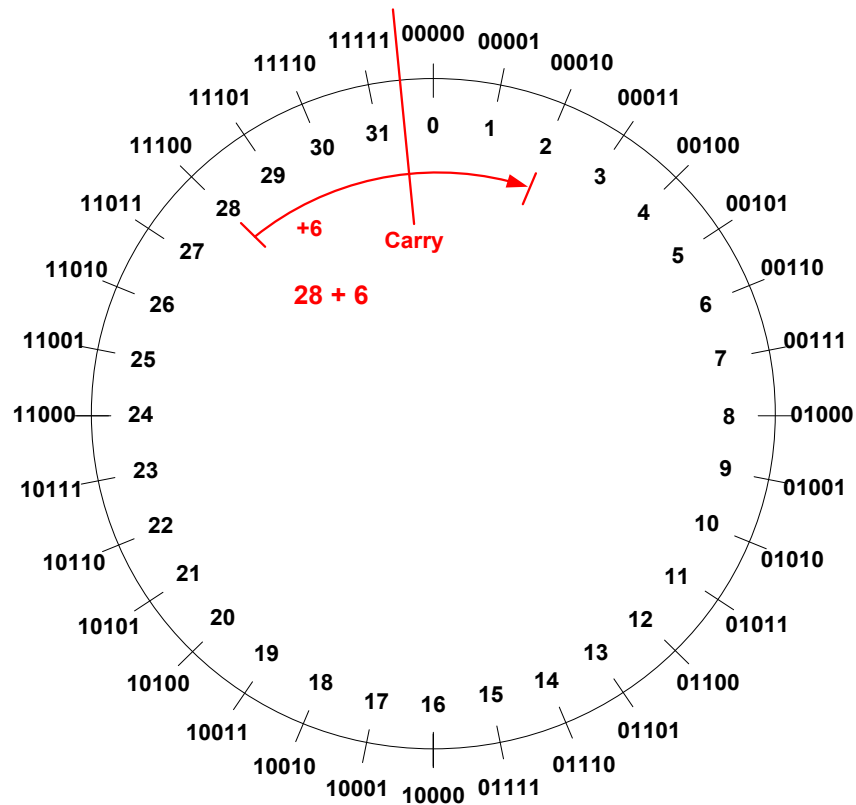
$$32_{10} = 100000_2$$

Carry/borrow flag bit for unsigned arithmetic

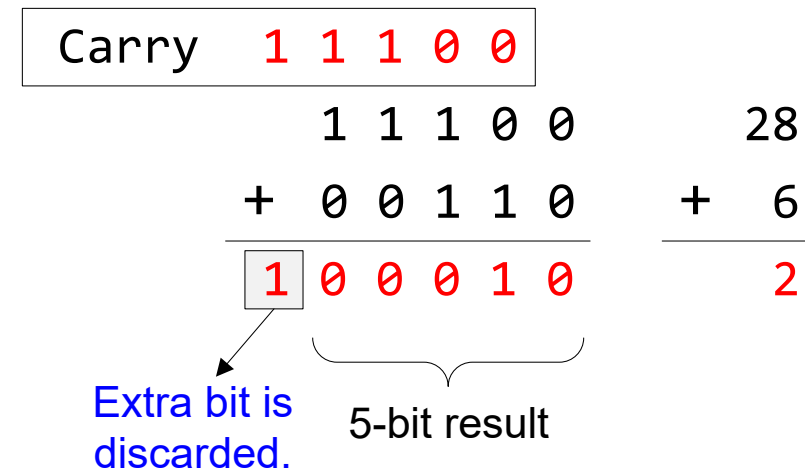
- Given **unsigned** integers a and b
- $c = a + b$
 - Carry happens if **c is too big to fit in n bits** (i.e., $c > 2^n - 1$).
- $c = a - b$
 - Borrow happens if $c < 0$.
- On ARM Cortex-M processors, the carry flag and the borrow flag are physically the same flag bit in the status register.
 - **For an unsigned subtraction, Carry = NOT Borrow**

Carry/borrow flag bit for unsigned numbers

If the traverse crosses the boundary between 0 and $2^n - 1 = 31$ ($n = 5$), the carry flag is set on addition and is cleared on subtraction.



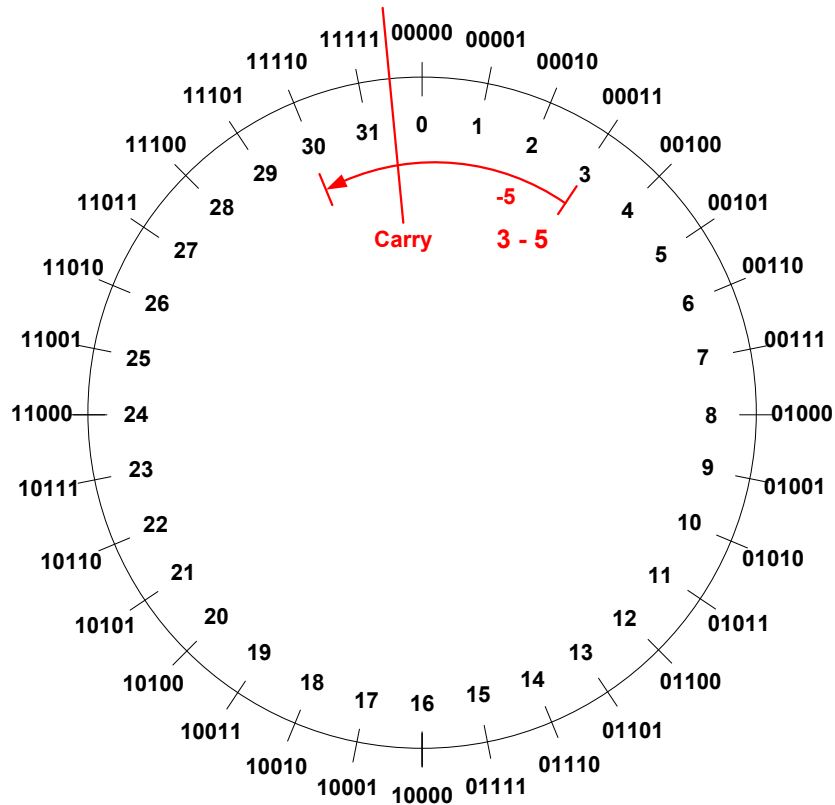
A carry occurs when adding 28 and 6



- Carry flag = 1, indicating carry has occurred on unsigned addition.
- Carry flag is 1 because the result crosses the boundary between 31 and 0.

Carry/borrow flag bit for unsigned numbers

If the traverse crosses the boundary between 0 and $2^n - 1$, the carry flag is set on addition and is cleared on subtraction.



Borrow	1	1	1	0	0	
	0	0	0	1	1	3
-	0	0	1	0	1	- 5
	1	1	1	1	0	30
5-bit result						

- Carry flag = 0, indicating borrow has occurred on unsigned subtraction.
- For subtraction, carry = NOT borrow.

A borrow occurs when subtracting 5 from 3.

Signed Integer Representation

- ▶ Three ways to represent signed binary integers:

- ▶ Signed magnitude

- ▶ $value = (-1)^{sign} \times Magnitude$

- ▶ One's complement ($\tilde{\alpha}$)

- ▶ $\alpha + \tilde{\alpha} = 2^n - 1$

- ▶ Two's complement ($\bar{\alpha}$)

- ▶ $\alpha + \bar{\alpha} = 2^n$

	Sign-and-Magnitude	One's Complement	Two's Complement
Range	$[-2^{n-1} + 1, 2^{n-1} - 1]$	$[-2^{n-1} + 1, 2^{n-1} - 1]$	$[-2^{n-1}, 2^{n-1} - 1]$
Zero	Two zeroes (± 0)	Two zeroes (± 0)	One zero
Unique Numbers	$2^n - 1$	$2^n - 1$	2^n

Signed Integers

Method 1: Signed Magnitude

Sign-and-Magnitude:

$$value = (-1)^{sign} \times Magnitude$$

- The most significant bit is the sign.
- The rest bits are magnitude.

▶ Example: in a 5-bit system

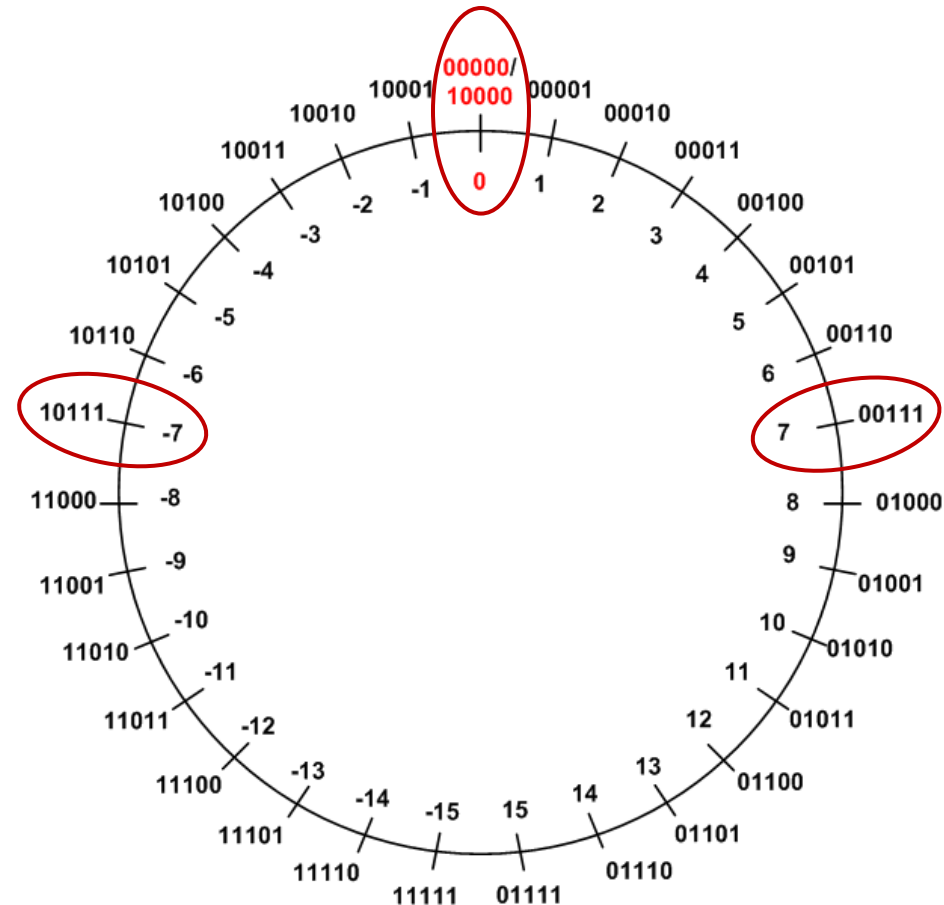
- ▶ $+7_{10} = 00111_2$
- ▶ $-7_{10} = 10111_2$

▶ Two ways to represent zero

- ▶ $+0_{10} = 00000_2$
- ▶ $-0_{10} = 10000_2$

▶ Not used in modern systems

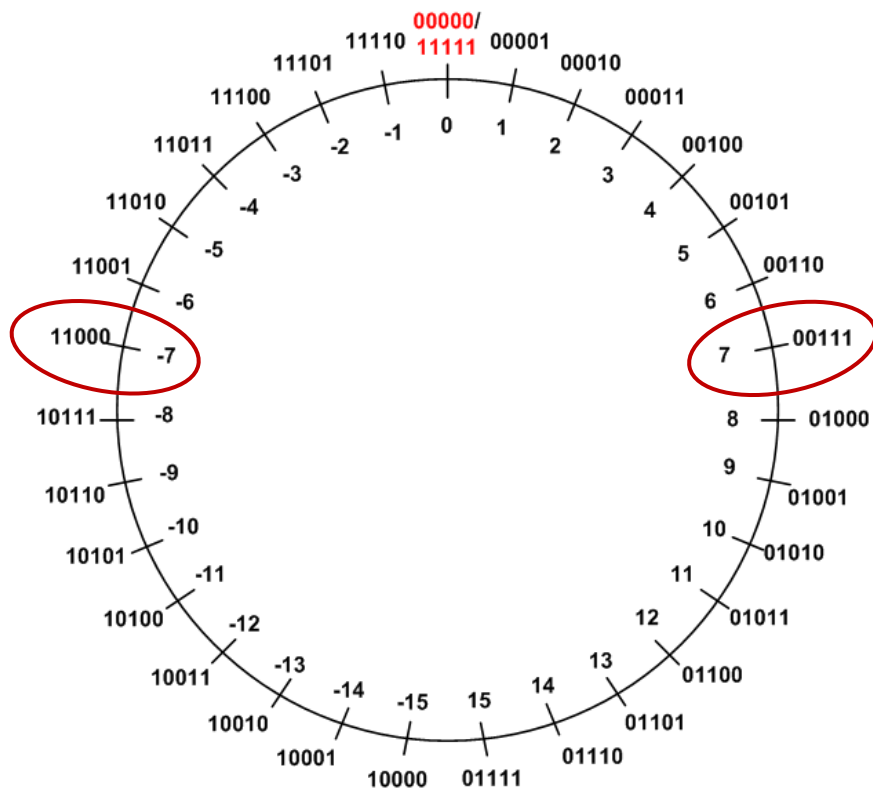
- ▶ Hardware complexity
- ▶ Two zeros



Signed Integers

Method 2: One's Complement

One's Complement ($\tilde{\alpha}$):
 $\alpha + \tilde{\alpha} = 2^n - 1$



The one's complement representation of a negative binary number is the bitwise NOT of its positive counterpart.

Example: in a 5-bit system

$$+7_{10} = 00111_2$$

$$-7_{10} = 11000_2$$

$$\begin{aligned} +7_{10} + (-7_{10}) &= 00111_2 + 11000_2 \\ &= 11111_2 \\ &= 2^5 - 1 \end{aligned}$$

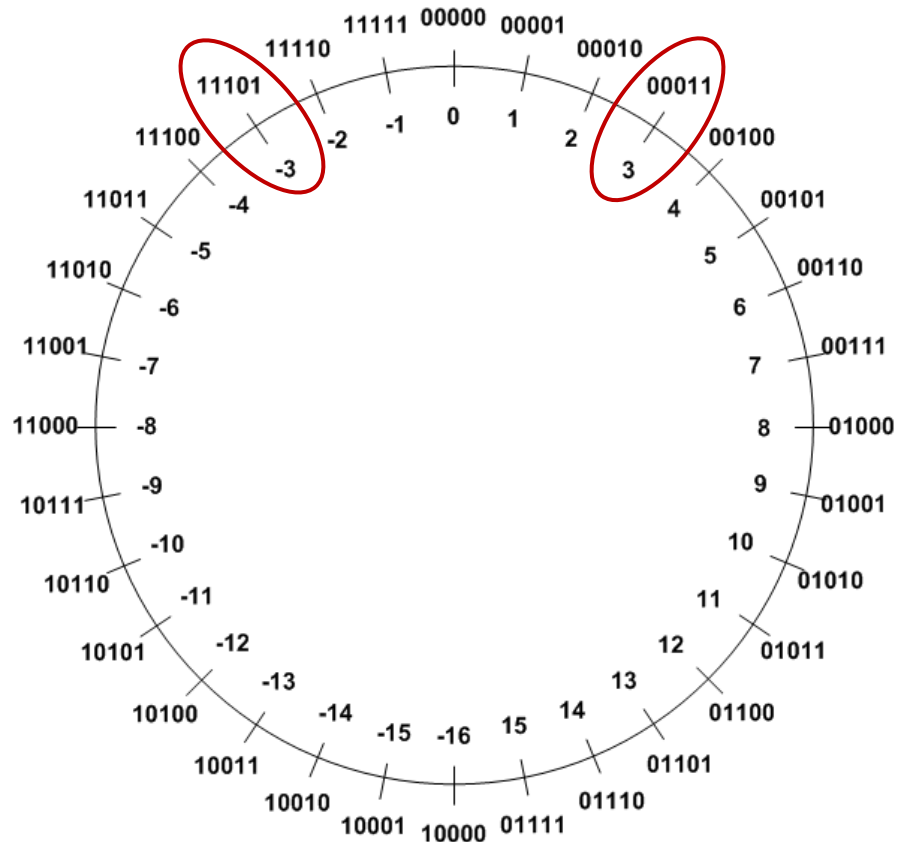


Signed Integers

Method 3: Two's Complement (TC)

Two's Complement ($\bar{\alpha}$):

$$\alpha + \bar{\alpha} = 2^n$$



TC of a negative number can be obtained by the bitwise NOT of its positive counterpart plus one.

Example 1: TC(3)

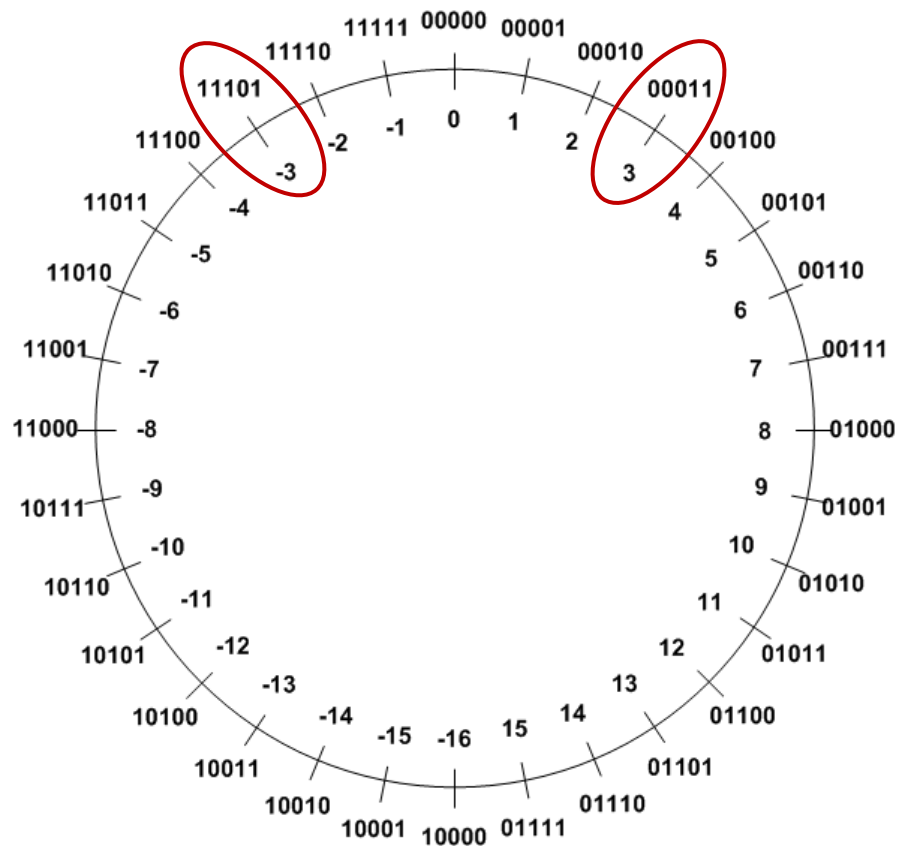
	Binary	Decimal
Original number	0b00011	3
Step 1: Invert every bit	0b11100	
Step 2: Add 1	+ 0b00001	
Two's complement	0b11101	-3

Signed Integers

Method 3: Two's Complement (TC)

Two's Complement (TC)

$$\alpha + \bar{\alpha} = 2^n$$



TC of a negative number can be obtained by the bitwise NOT of its positive counterpart plus one.

Example 2: **TC(-3)**

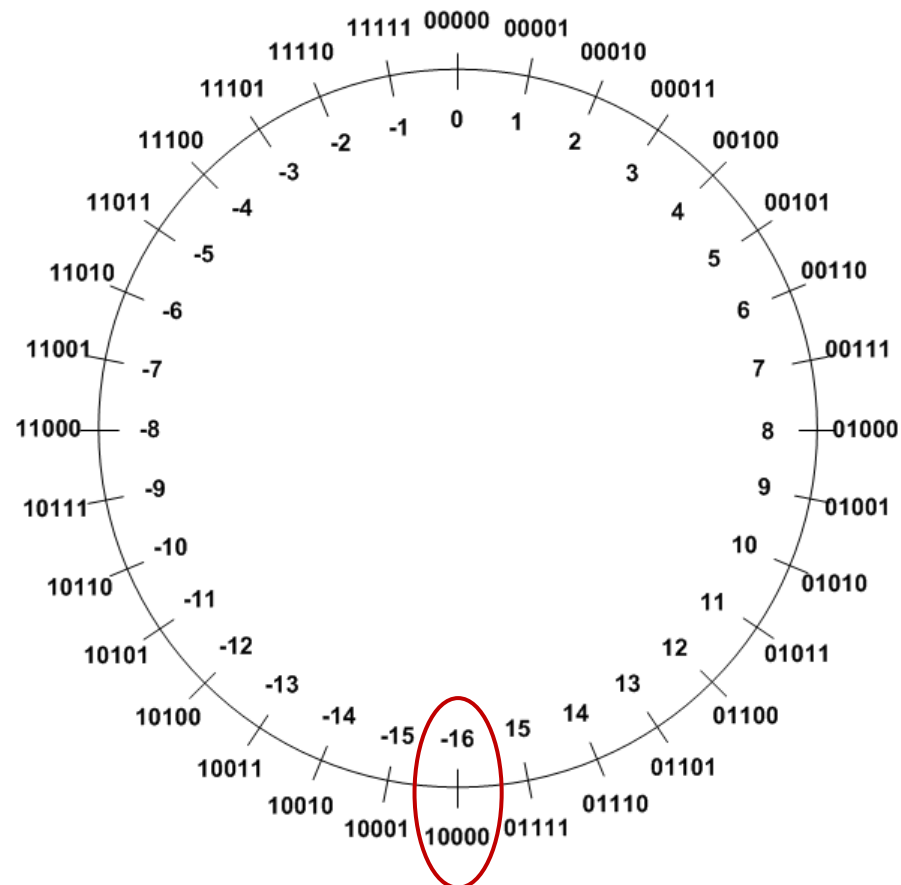
	Binary	Decimal
Original number	0b11101	-3
Step 1: Invert every bit	0b00010	
Step 2: Add 1	+ 0b00001	
Two's complement	0b00011	3

Signed Integers

Method 3: Two's Complement (TC)

Two's Complement (TC)

$$\alpha + \bar{\alpha} = 2^n$$



TC of a negative number can be obtained by the bitwise NOT of its positive counterpart plus one.

Example 3: **TC(-16)**

	Binary	Decimal
Original number	10000	-16
Step 1: Invert every bit	01111	
Step 2: Add 1	+ 10000	
Two's complement	10000	-16

Negation of -16 in 5-bit two's complement wraps back to itself, meaning the most negative number's two's complement is itself. (Number range is [-16, 15], so 16 is out of range)

Quiz

- ▶ Calculate TC(-16) for a 6-bit system



Quiz ANS

- ▶ Calculate TC(-16) for a **6-bit** system
- ▶ For a 6-bit two's complement number: the range of representable integers is from -32 to $+31$.
- ▶ -16 in 6-bit two's complement is 110000
 - ▶ Write 16 in binary: 010000
 - ▶ Take the two's complement (invert bits and add 1):
 - ▶ Invert bits: 101111, Add 1: $101111 + 1 = 110000$
- ▶ To take the negation of this (i.e., find the two's complement of 110000):
 - ▶ Invert bits: 001111, Add 1: $001111 + 1 = 010000$
 - ▶ This is 16 in binary, so the negation of -16 is $+16$ as expected
- ▶ Unlike the 5-bit case where -16 is the minimum and its negation wraps onto itself, in 6 bits -16 behaves normally with correct negation
- ▶ TC(-32) for a 6-bit system is itself (since $+32$ is out of range of $[-32, 31]$)



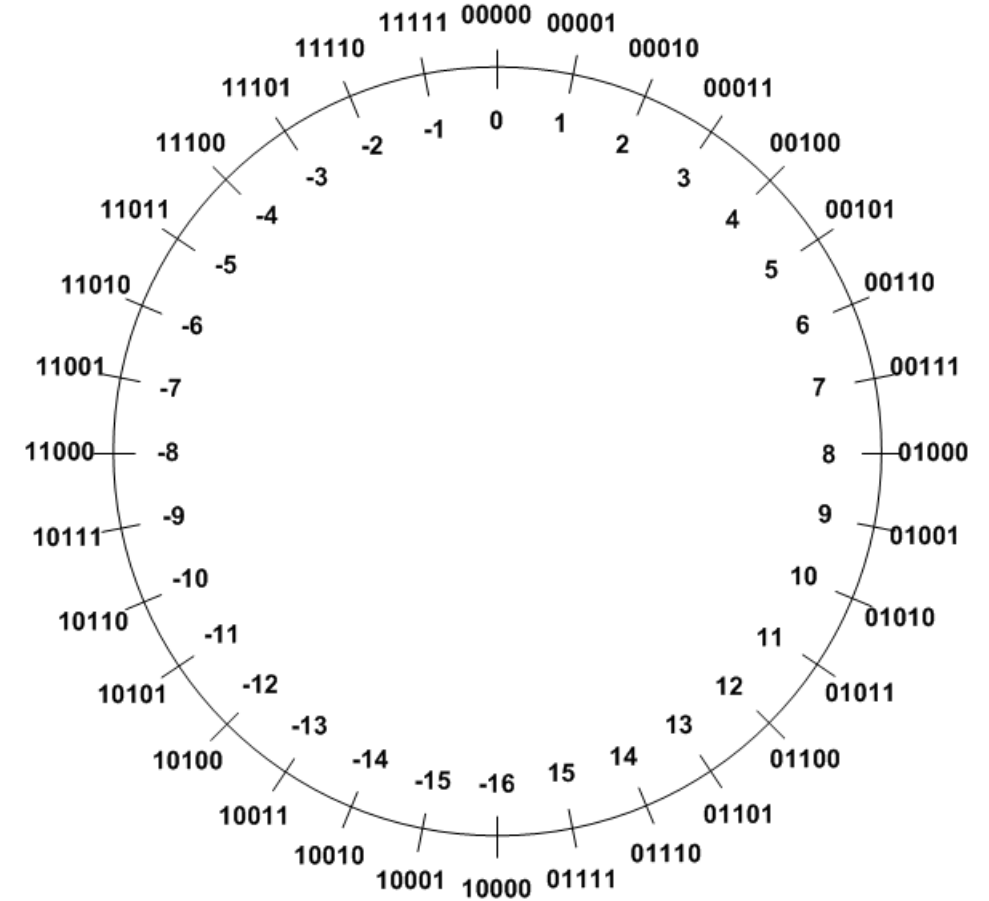
Two's Complement (TC)

- ▶ Two's complement gets its name from the rule that “*the unsigned sum of an n -bit number and its n -bit negative is 2^n* ”; hence, the negation or complement of a number x is $2^n - x$, or its “Two's complement”
- ▶ Signed arithmetic:
 - ▶ For 5-bit system, signed number $x = 00011 = 3$, $TC(x) = 11101 = -3$, so their **signed sum** is $00011 + 11101 = 00000$ (in decimal $3 + (-3) = 0$)
- ▶ Unsigned arithmetic:
 - ▶ Unsigned number $00011=3$, $11101=29$, so their **unsigned sum** is $00011 + 11101 = 00000$ (in decimal $3 + 29 = 32 = 2^5$). The result is incorrect as Carry flag = 1: 32 cannot be represented in 5 bits since it exceeds the largest unsigned value of $2^5 - 1$

Two's Complement for 8-bit System

8-bit signed Int (Two's Complement)	8-bit unsigned Int	Binary
-128	128	1 0 0 0 0 0 0 0
-127	129	1 0 0 0 0 0 0 1
...	...	...
-2	254	1 1 1 1 1 1 1 0
-1	255	1 1 1 1 1 1 1 1
0	0	0 0 0 0 0 0 0 0
1	1	0 0 0 0 0 0 0 1
...	...	...
127	127	0 1 1 1 1 1 1 1

Note: Most significant bit (MSB) is the sign bit for signed int



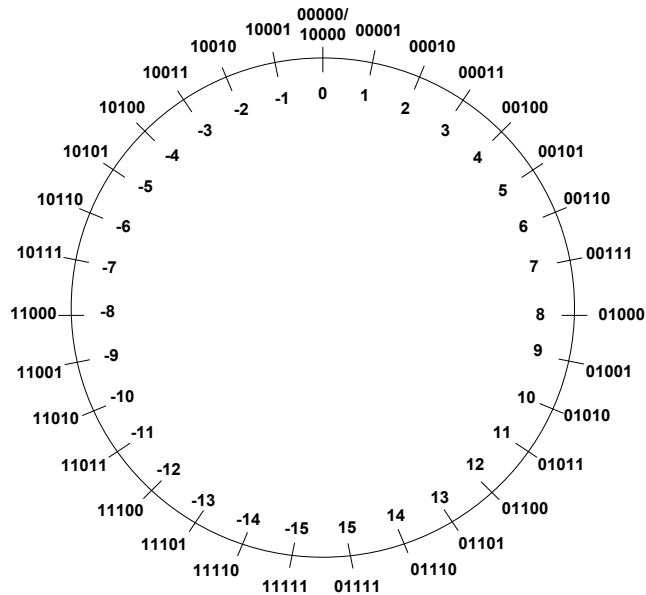
Sign Extension

Decimal	Binary		
	4-bit	8-bit	32-bit
3_{ten}	0011_{two}	$0000\ 0011_{\text{two}}$	$0000\ 0000\ 0000\ 0011_{\text{two}}$
-3_{ten}	1101_{two}	$1111\ 1101_{\text{two}}$	$1111\ 1111\ 1111\ 1101_{\text{two}}$

- Sign extension for unsigned int: fill in 0's from the left
- Sign extension for signed int: fill in the sign bit from the left



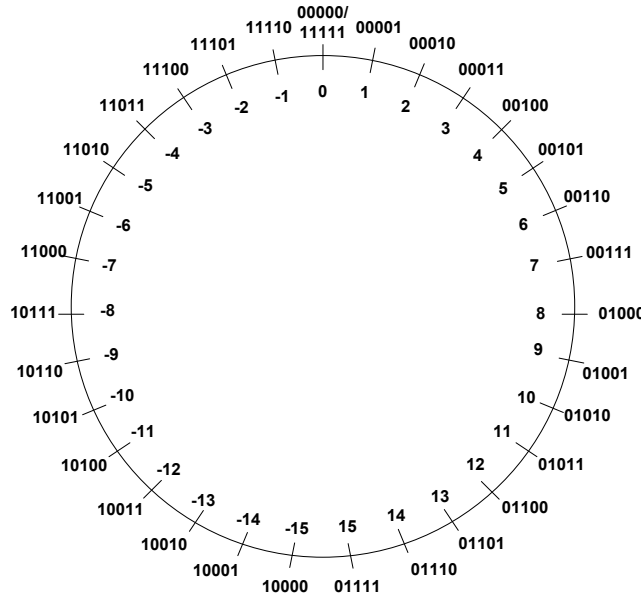
Comparison



Signed magnitude
representation

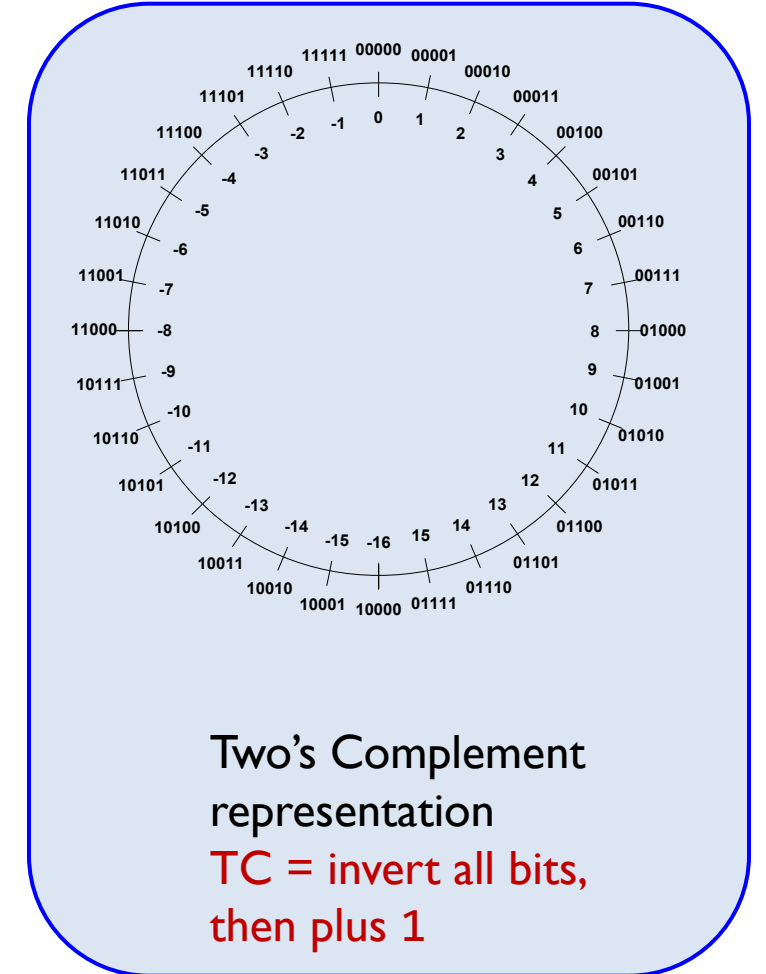
0 = positive

1 = negative



One's complement
representation

Negative = invert all
bits of a positive

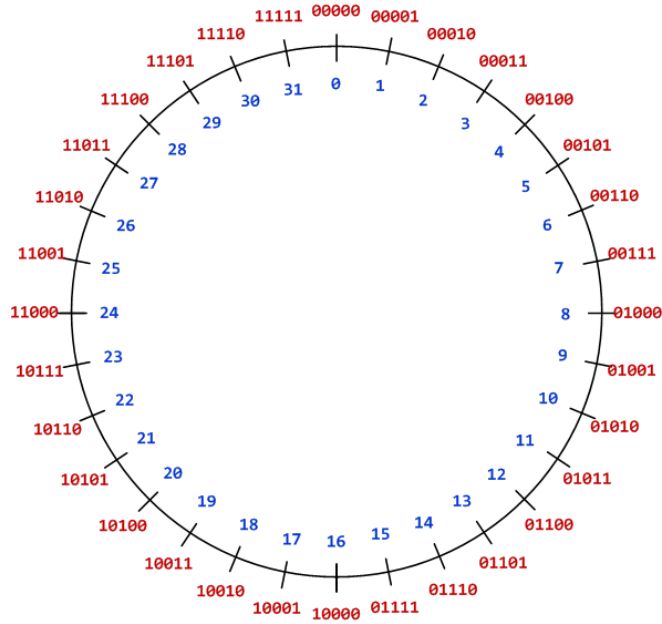


Two's Complement
representation

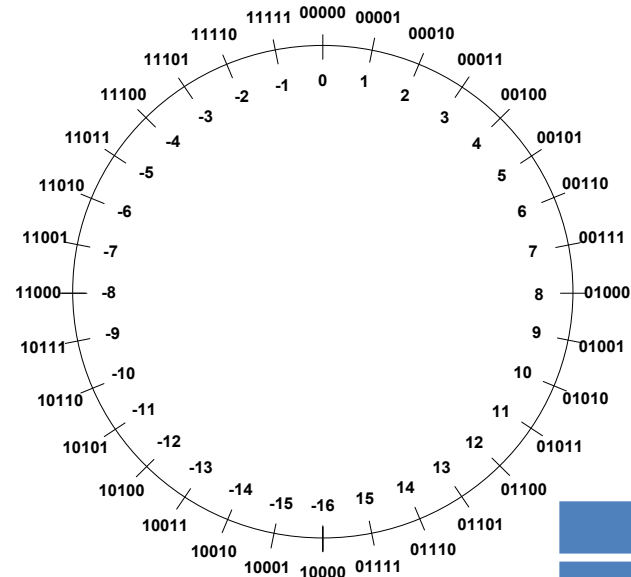
TC = invert all bits,
then plus 1

Used in modern computers!

Comparison: unsigned vs. signed



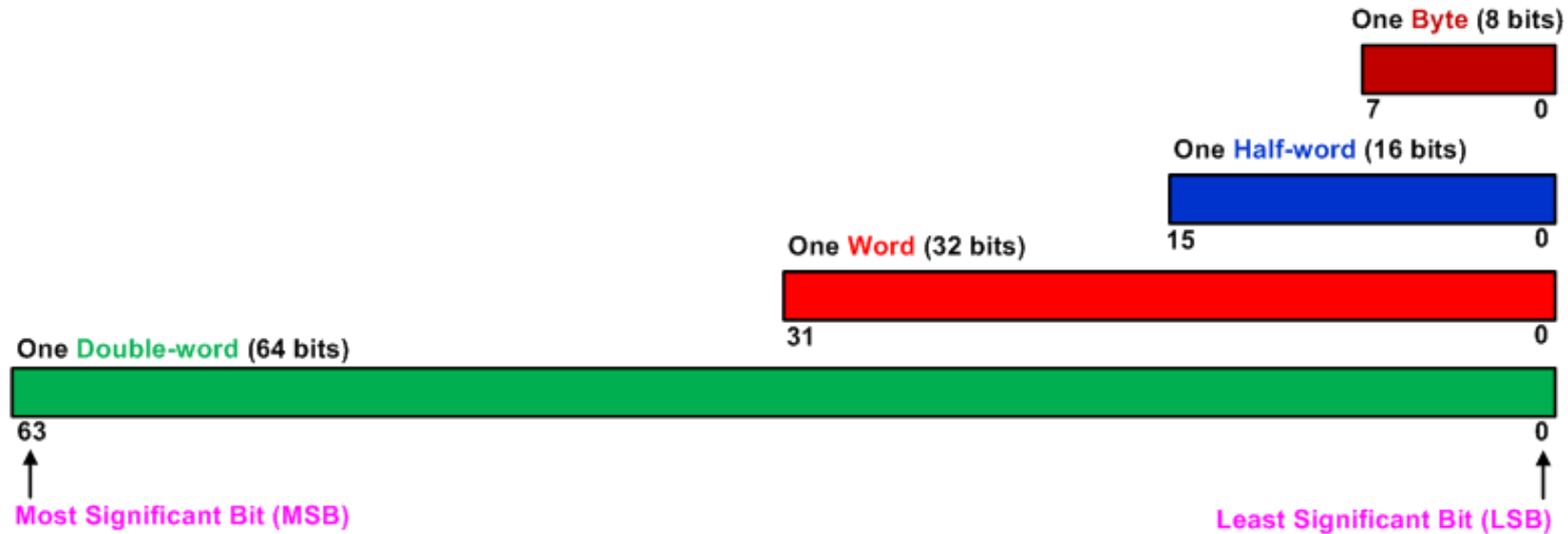
Unsigned int
representation
Range $[0, 31]$



Two's Complement
representation
Range $[-16, 15]$
TC = invert all bits,
then add 1

	Unsigned	Two's Complement Signed
Range	$[0, 2^n - 1]$	$[-2^{n-1}, 2^{n-1} - 1]$
Zero	One zero	One zero
Unique Numbers	2^n	2^n

Range of Signed Integers (Two's Complement)



Storage Size	Range	Powers of 2
Signed Byte	-128 to +127	-2^7 to 2^7-1
Signed half-word	-32,768 to +32,767	-2^{15} to $2^{15}-1$
Signed Word	-2,147,483,648 to +2,147,483,647	-2^{31} to $2^{31}-1$
Signed Double-word	-9,223,372,036,854,775,808 to +9,223,372,036,854,775,807	-2^{63} to $2^{63}-1$

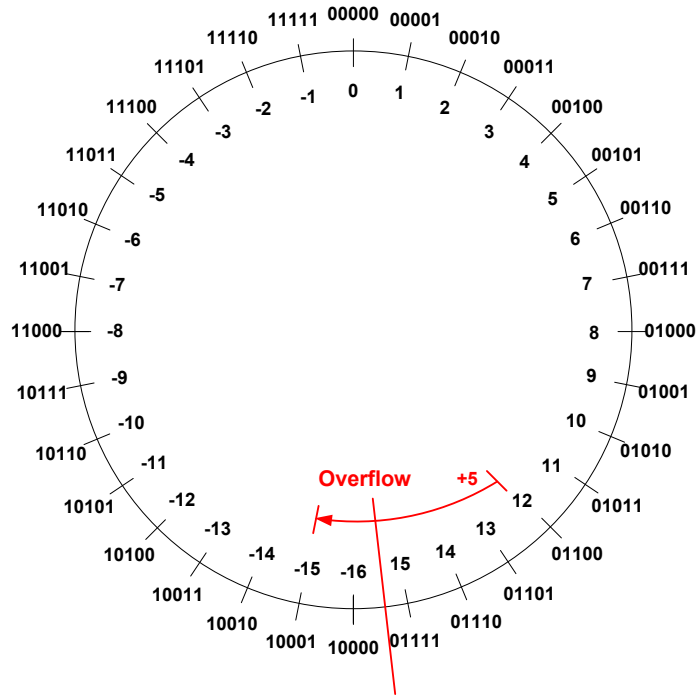
Overflow Flag for Signed Arithmetic

- ▶ When adding signed numbers represented in two's complement, overflow occurs only in two scenarios:
 1. adding two positive numbers but getting a non-positive result, or
 2. adding two negative numbers but yielding a non-negative result.
- ▶ Similarly, when subtracting signed numbers, overflow occurs in two scenarios:
 1. subtracting a positive number from a negative number but getting a positive result, or
 2. subtracting a negative number from a positive number but producing a negative result.
- ▶ Overflow cannot occur when adding operands with different signs or when subtracting operands with the same signs.
 - ▶ Why?

Overflow Flag for Signed Arithmetic

- ▶ Overflow cannot occur when adding 2 operands with different signs or when subtracting 2 operands with the same sign. Proof:
- ▶ A n -bit signed int has the range $[-2^{n-1}, 2^{n-1}-1]$
 - ▶ $n = 5$, number range $[-16, 15]$
- ▶ 2 operands with different signs: positive one in the range of $[0, 2^{n-1}-1]$, negative one in the range of $[-2^{n-1}, -1]$. So the range of their sum must be $[0-2^{n-1}, 2^{n-1}-1+(-1)] = [-2^{n-1}, 2^{n-1}-2] \in [-2^{n-1}, 2^{n-1}-1] \rightarrow$ No overflow!
 - ▶ Positive number range $[0, 15]$, negative number range $[-16, -1]$. Range of their sum $[0-16, 15-1] = [-16, 14]$
- ▶ 2 operands with the same sign: if both are positive and in the range of $[0, 2^{n-1}-1]$, then the range of their difference must be $[0-(2^{n-1}-1), 2^{n-1}-1-0] = [-(2^{n-1}-1), 2^{n-1}-1]$; if both are negative and in the range of $[-2^{n-1}, -1]$, then the range of their difference must be $[-2^{n-1}-(-1), -1-(-2^{n-1})] = [-2^{n-1}+1, 2^{n-1}-1] \in [-2^{n-1}, 2^{n-1}-1] \rightarrow$ No overflow!
 - ▶ Both positive numbers $[0, 15]$, range of difference $[0-15, 15-0] = [-15, 15]$
 - ▶ Both negative numbers $[-16, -1]$, range of difference $[-16-(-1), -1-(-16)] = [-15, 15]$

Overflow for Signed Add

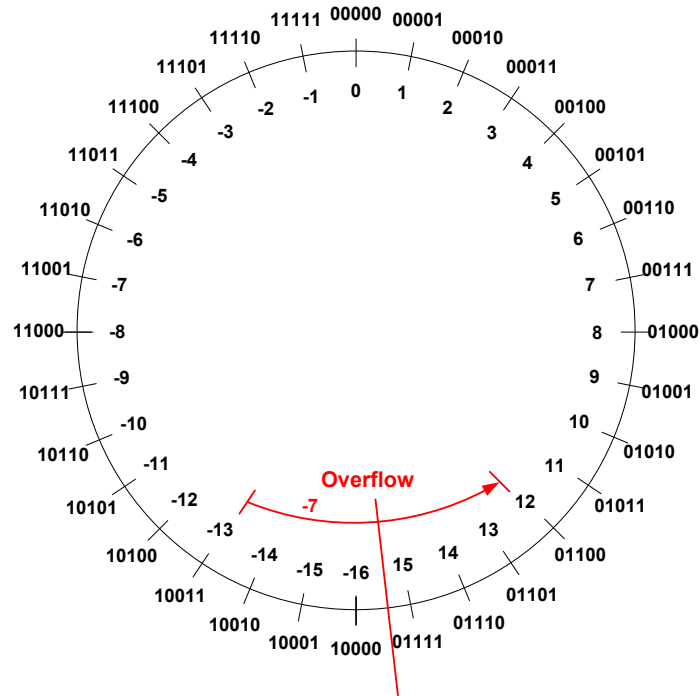


Overflow occurs when adding two positive integers but getting a negative result.

0 1 1 0 0	12
+ 0 0 1 0 1	+ 5
1 0 0 0 1	-15
5-bit result	

1. On addition, overflow occurs if $sum \geq 2^4$ when adding two positives.
2. Overflow never occurs when adding two numbers with different signs.

Overflow for Signed Add



Overflow occurs when adding two negative integers but getting a positive result.

	1	0	0	1	1		-13
+	1	1	0	0	1		+ -7
	1	0	1	1	0	0	12

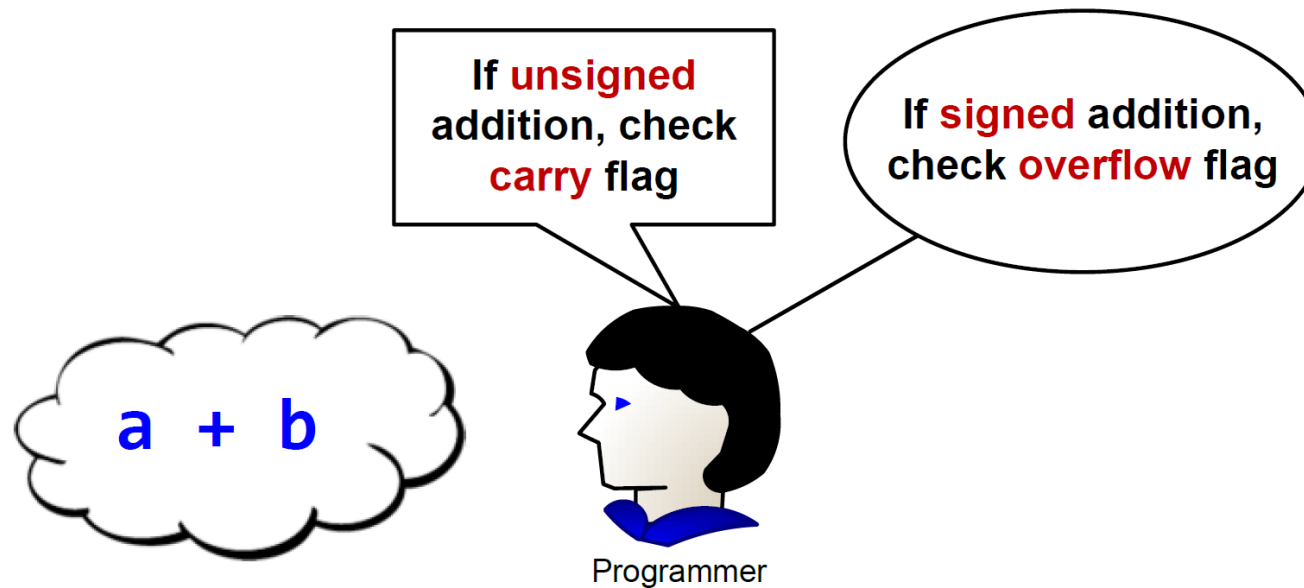
Extra bit is discarded.

5-bit result

On addition, overflow occurs if $sum < -2^4$ when adding two negatives.

Signed or unsigned

- ▶ Whether the carry flag or the overflow flag should be used depends on the programmer's intention.



- ▶ When programming in high-level languages such as C, the compiler automatically chooses to use the carry or overflow flag based on how this integer is declared in source code ("**int**" or "**unsigned int**").

Signed or Unsigned (5-bit System)

$a = 0b10000$

$b = 0b10000$

$c = a + b$

- Whether the carry flag or the overflow flag should be used depends on the programmer's intention: Are a and b signed or unsigned numbers?

```
uint a;  
uint b;  
...  
c = a + b  
...
```

Check the carry flag!

C Program

```
int a;  
int b;  
...  
c = a + b  
...
```

Check the overflow flag!

C Program

Signed or Unsigned (5-bit System)

$a = 0b10000$

$b = 0b10000$

$c = a + b$

- ▶ Are a and b signed or unsigned numbers?
- ▶ CPU does not know and does not care; it sets up both carry flag and overflow flags.
 - ▶ $10000 + 10000 = 00000$, and carry/overflow flags.
- ▶ It is software's (programmer/compiler) responsibility to interpret the flags.
 - ▶ The C compiler uses either the carry or the overflow flag based on how this integer is declared in source code ("uint" or "int").

If unsigned:

uint $a, b;$

$a = 16$

$b = 16$

$c = a + b$

$= 32 > 2^5 - 1$

Carry flag set

If signed:

int $a, b;$

$a = -16$

$b = -16$

$c = a + b$

$= -32 < -2^4$

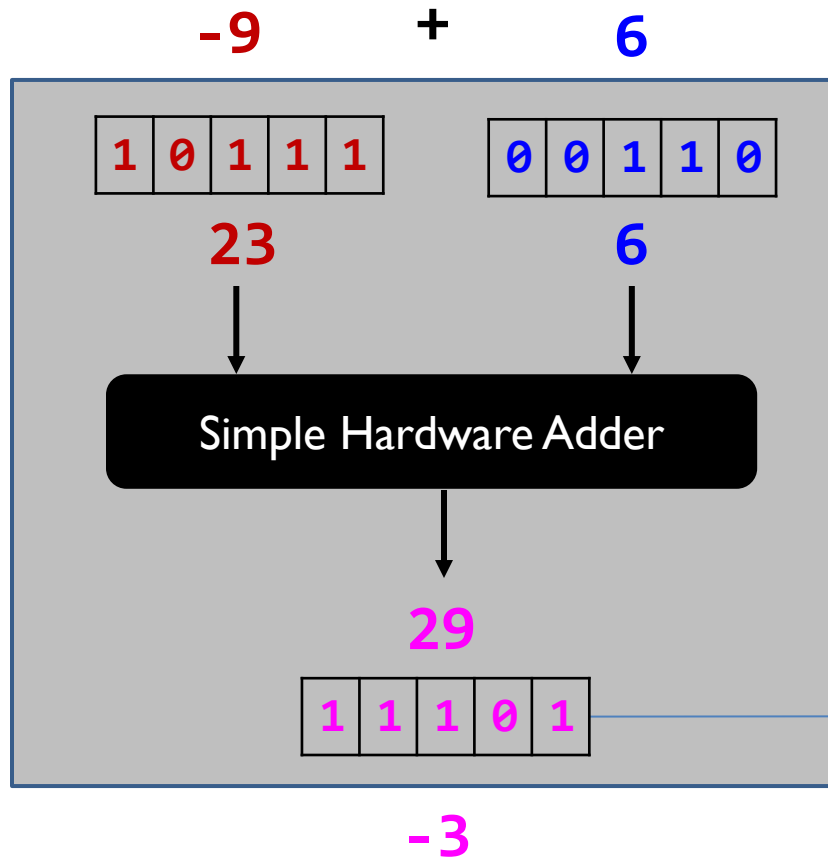
Overflow flag set

Two's Complement Simplifies Hardware Implementation

- ▶ In two's complement, **the same hardware** works correctly for both signed and unsigned addition/subtraction.
- ▶ If the product is required to keep the same number of bits as operands, unsigned multiplication hardware works correctly for signed numbers.
- ▶ However, this is not true for division. (not discussed in this course)

Operation	Are signed and unsigned operations the same?
Addition	Yes
Subtraction	Yes
Multiplication	Yes if the product is required to keep the same number of bits as operands
Division	No

Adding two signed integers: $(-9) + 6$

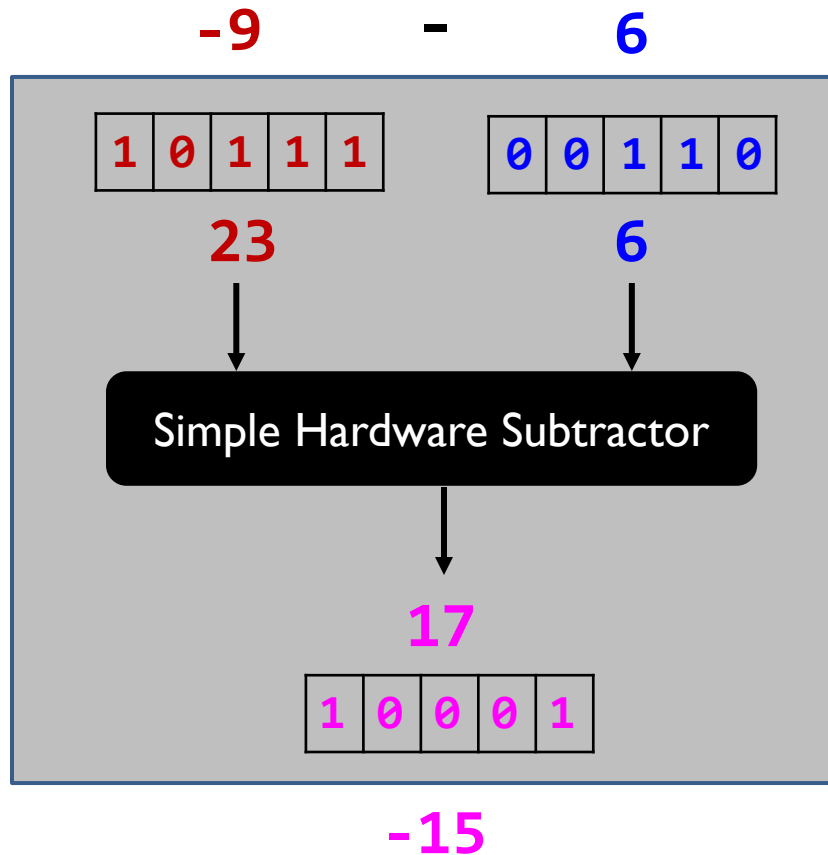


9
01001 → flip → **10110** → +1 → **10111**
-9
Two's Complement

- ▶ Same binary arithmetic: $10111 + 00110 = 11101$ can represent:
- ▶ Either signed arithmetic : $(-9) + 6 = -3$
 - ▶ Or unsigned arithmetic : $23 + 6 = 29$

flip → **00010** → +1 → **00011**
3
Two's Complement Counterpart

Subtracting two signed integers: (-9) - 6



$$\begin{array}{c} 9 \\ 01001 \end{array} \xrightarrow{\text{flip}} 10110 \xrightarrow{+1} \begin{array}{c} -9 \\ 10111 \end{array}$$

Two's Complement

- Same binary arithmetic: $10111 - 00110 = 10001$ can represent:
- Either signed arithmetic : $(-9) - 6 = -15$
 - Or unsigned arithmetic : $23 - 6 = 17$

$$\begin{array}{c} 15 \\ 01111 \end{array} \xrightarrow{\text{flip}} 10000 \xrightarrow{+1} \begin{array}{c} -15 \\ 10001 \end{array}$$

Condition Codes

Bit	Name	Meaning after add or sub
N	negative	Result's MSB = 1 (signed result < 0)
Z	zero	Result == 0
C	carry	Unsigned arithmetic out of range
V	overflow	Signed arithmetic out of range



CPSR (Current Program Status Register)

- C is set upon an **unsigned** addition if the result is out of range
- C is cleared upon an **unsigned** subtract if the result is out of range
- V is set upon a **signed** addition or subtraction if the result is out of range

Why do we care about these bits?

Carry and Overflow Flags

Carry flag C = 1 (Borrow flag = 0) upon an unsigned addition if the result is out of range (true result > 2ⁿ-1)
 Carry flag C = 0 (Borrow flag = 1) upon an unsigned subtraction if the result is out of range (true result < 0)
 Overflow flag V = 1 upon a signed addition or subtraction if the result is out of range (true result > 2ⁿ⁻¹-1 or true result < -2ⁿ⁻¹)

$$c = a \pm b$$

	Carry (for unsigned)	Overflow (for signed)
Add	<i>Carry = 1</i> if <i>c</i> is too large to fit in.	<i>Overflow = 1</i> if <i>c</i> is too large or too small to fit in
Subtract	<i>Borrow = 1</i> , i.e. <i>Carry = 0</i> if <i>a < b</i> .	

- ARM Cortex-M has no dedicated borrow flag, carry flag is reused.
 - For unsigned subtract, $Borrow = \overline{Carry}$

- Signed Subtraction is converted to sign addition
 - $a - b = a + (-b)$

Characters

American
Standard
Code for
Information
Interchange

- Standard ASCII
0 - 127
- Extended ASCII
0 - 255
- ANSI
0 - 255
- Unicode
0 - 65535

Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char
0	00	NUL	32	20	SP	64	40	@	96	60	`
1	01	SOH	33	21	!	65	41	A	97	61	a
2	02	STX	34	22	"	66	42	B	98	62	b
3	03	ETX	35	23	#	67	43	C	99	63	c
4	04	EOT	36	24	\$	68	44	D	100	64	d
5	05	ENQ	37	25	%	69	45	E	101	65	e
6	06	ACK	38	26	&	70	46	F	102	66	f
7	07	BEL	39	27	'	71	47	G	103	67	g
8	08	BS	40	28	(	72	48	H	104	68	h
9	09	HT	41	29	)	73	49	I	105	69	i
10	0A	LF	42	2A	*	74	4A	J	106	6A	j
11	0B	VT	43	2B	+	75	4B	K	107	6B	k
12	0C	FF	44	2C	,	76	4C	L	108	6C	l
13	0D	CR	45	2D	-	77	4D	M	109	6D	m
14	0E	SO	46	2E	.	78	4E	N	110	6E	n
15	0F	SI	47	2F	/	79	4F	O	111	6F	o
16	10	DLE	48	30	0	80	50	P	112	70	p
17	11	DC1	49	31	1	81	51	Q	113	71	q
18	12	DC2	50	32	2	82	52	R	114	72	r
19	13	DC3	51	33	3	83	53	S	115	73	s
20	14	DC4	52	34	4	84	54	T	116	74	t
21	15	NAK	53	35	5	85	55	U	117	75	u
22	16	SYN	54	36	6	86	56	V	118	76	v
23	17	ETB	55	37	7	87	57	W	119	77	w
24	18	CAN	56	38	8	88	58	X	120	78	x
25	19	EM	57	39	9	89	59	Y	121	79	y
26	1A	SUB	58	3A	:	90	5A	Z	122	7A	z
27	1B	ESC	59	3B	;	91	5B	[	123	7B	{
28	1C	FS	60	3C	<	92	5C	\	124	7C	
29	1D	GS	61	3D	=	93	5D	]	125	7D	}
30	1E	RS	62	3E	>	94	5E	^	126	7E	~
31	1F	US	63	3F	?	95	5F	_	127	7F	DEL

Standard ASCII: Encoding 128 characters

Null-terminated String

```
char str[13] = "ARM Assembly";  
// The length has to be at least  
// 13 even though it has 12  
// letters. The NULL terminator  
// should be included.
```

or simply

```
char str[] = "ARM Assembly";
```

Memory Address	Memory Content	Letter
str + 12 →	0x00	\0
str + 11 →	0x79	y
str + 10 →	0x6C	l
str + 9 →	0x62	b
str + 8 →	0x6D	m
str + 7 →	0x65	e
str + 6 →	0x73	s
str + 5 →	0x73	s
str + 4 →	0x41	A
str + 3 →	0x20	space
str + 2 →	0x4D	M
str + 1 →	0x52	R
str →	0x41	A

String Comparison

Strings are compared based on their ASCII values

- ▶ “j” < “jar” < “jargon” < “jargonize”
- ▶ “CAT” < “Cat” < “DOG” < “Dog” < “cat” < “dog”
- ▶ “12” < “123” < “2” < “AB” < “Ab” < “ab” < “abc”

String Length

- ▶ Strings are terminated with a **null** character (NUL, ASCII value 0x00)

Pointer dereference operator *

```
int strlen (char *pStr){
    int i = 0;

    // loop until *pStr is NULL
    while( *pStr ) {
        i++;
        pStr++;
    }
    return i;
}
```

Array subscript operator []

```
int strlen (char *pStr){
    int i = 0;

    // loop until pStr[i] is NULL
    while( pStr[i] )
        i++;

    return i;
}
```

Convert to Upper Case

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
41	42	43	44	45	46	47	48	49	4A	4B	4C	4D	4E	4F	50	51	52	53	54	55	56	57	58	59	5A

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
61	62	63	64	65	66	67	68	69	6A	6B	6C	6D	6E	6F	70	71	72	73	74	75	76	77	78	79	7A

$$\text{'a'} - \text{'A'} = 0x61 - 0x41 = 0x20 = 32$$

Pointer dereference operator *

```
void toUpper(char *pStr){
    char *p = pStr;
    for(; *p; ++p){
        if(*p >= 'a' && *p <= 'z')
            *p -= 'a' - 'A';
            //or: *p -= 32;
    }
}
```

Array subscript operator []

```
void toUpper(char *pStr){
    int i;
    char c = pStr[0];
    for(i = 0; c; i++, c = pStr[i];) {
        if(c >= 'a' && c <= 'z')
            pStr[i] -= 'a' - 'A';
            // or: pStr[i] -= 32;
    }
}
```

Summary

- ▶ Unsigned integer arithmetic
- ▶ Signed integer arithmetic
 - ▶ 2's complement
- ▶ Carry and Overflow Flags
 - ▶ Carry answers: “Did unsigned math overflow?”
 - ▶ Overflow answers: “Did signed math overflow?”
 - ▶ Hardware sets both. Software decides which one is used
- ▶ ASCII strings

References

- ▶ Lecture 1: Why use two's complement?
 - ▶ <https://www.youtube.com/watch?v=IJCefqV80ck&list=PLRJhV4hUhlymmp5CCelFPyxbknsdcXCc8&index=1>
- ▶ Lecture 2: Carry flag for unsigned addition and subtraction
 - ▶ <https://www.youtube.com/watch?v=MxGW2WurKuM&list=PLRJhV4hUhlymmp5CCelFPyxbknsdcXCc8&index=2>
- ▶ Lecture 3: Overflow flag for signed addition and subtraction
 - ▶ <https://www.youtube.com/watch?v=BlN6iyYIGio&list=PLRJhV4hUhlymmp5CCelFPyxbknsdcXCc8&index=3>

